EXPERIMENT 08

Configuring User Management and Role-Based Access Control (RBAC) in Zoho Creator Application

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SUBMITTED BY

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1. Aim

To examine how user management and role-based access control (RBAC) are implemented in Zoho Creator by studying the Permissions, Roles and Data Sharing settings of the Loan Management System, and to understand how these three layers together determine which users can view and modify loan records.

2. Objectives

- To start from the Zoho Creator home page and the application dashboard, which serve as the entry points to the Loan Management System.
- To study the permission sets defined in the application: Branch Manager, Credit Manager, Customer Service Executive, Loan Officer, Read and Write.
- To build and review the role hierarchy, from the CEO at the top down to the operational roles.
- To create a data sharing rule that gives one role access to records submitted by another role.
- To save the rule and confirm that every field was stored exactly as it was configured.

3. Software and Tools Required

- Zoho Creator, the web-based low-code application development platform
- The Loan Management System application
- A web browser with an active internet connection
- A Zoho account with access to the application in Edit mode

4. Introduction and Theory

4.1 Role-Based Access Control in Zoho Creator

Role-based access control grants access according to the job a person performs, not according to who the individual is. Zoho Creator applies this principle through three layers that operate together. Permissions determine what a user may do with the forms, reports and pages of an application. Roles place users in a hierarchy that determines whose records a user is able to see. Data sharing rules then open specific records to other roles or users, beyond what the hierarchy alone would allow.

4.2 Permissions, Roles and Data Sharing

Permission set. A permission set, also called a profile, is a named collection of rights, for example the right to view, add, edit or delete records in selected forms.

Role. A role is a position in the organisation tree. A user who holds a role can normally see the records belonging to the roles below it.

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Data sharing rule. A data sharing rule is an explicit exception to the hierarchy. It names a form, the role or users whose records are being shared, the role or users who receive access, and the level of access granted, which is either Read or Read/Write.

Working together, the three layers allow an application to present each user with only the loan data that their work requires.

5. Key Concepts

Concept	Description
Permission Set	A named collection of rights that defines what a user can do with the forms, reports and pages of the application.
Role	A position in the role hierarchy that determines whose records a user is able to see.
Role Hierarchy	A tree of roles in which each role can see the records of the roles beneath it.
Data Sharing Rule	A rule that shares the records of a chosen form from one role or user group with another, at Read or Read/Write access.
Level of Access	The degree of access given by a sharing rule: Read allows records to be viewed, while Read/Write allows them to be viewed and edited.

6. Application Specifications

Parameter	Detail
Application Name	Loan Management System
Primary Forms	Borrower, Loan Application, Loan Type, Repayment
Permission Sets	Branch Manager, Credit Manager, Customer Service Executive, Loan Officer, Read, Write
Role Hierarchy	CEO → Branch Manager → Credit Manager, Loan Officer, Customer Service Executive
Data Sharing Rule	Customer_Service_to_Branch_Manager on the Loan Application form (Read/Write)

7. Procedure

The figures in this record are illustrative representations drawn with sample values. They are not screenshots of the application.

Step 1: Open the Zoho Creator Home Page

Signing in to Zoho Creator opens the Home page, which lists all applications available in the account. The Loan Management System appears here as an application card.

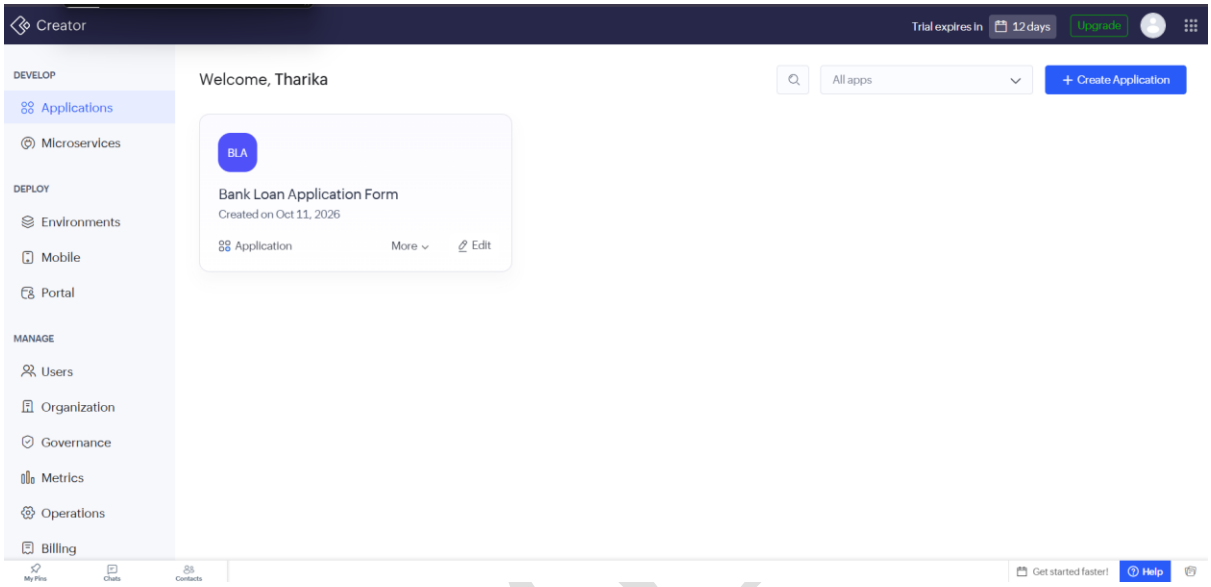


Figure 8.1: Zoho Creator Home - Loan Management System (Illustrative Representation, Not a Screenshot)

As Figure 8.1 shows, the card for the Loan Management System sits alongside the Application and Edit shortcuts.

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Step 2: Open the Dashboard

Opening the application loads its Dashboard, which presents summary tiles and charts built from the loan data.



Figure 8.2: Loan Management System - Dashboard (Access Mode) (Illustrative Representation, Not a Screenshot)

The dashboard in Figure 8.2 contains four summary tiles (Borrowers, Loan Applications, Loan Types and Repayments), a Distribution by Loan Status pie chart, Review Status counts, a Quick Links panel, and a Distribution by Created By bar chart.

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Step 3: Enter Edit Mode and Open Settings

Selecting Edit this application switches the application to Edit mode. From the top navigation bar, which offers Design, Workflow and Settings, the Settings tab is then selected.

Step 4: Open User Permissions

Within the PERMISSIONS group, User Permissions is selected. This opens the Users page, which carries three tabs: Permissions, Roles and Data Sharing. The Permissions tab lists six permission sets, each with a short description and an Add Permission option, as summarised below.

Permission Set	Purpose
Branch Manager	Monitors loan application statuses, including pending, approved and rejected applications.
Credit Manager	Reviews assigned applications and updates their review status as required.
Customer Service Executive	Captures new borrower profiles and initial loan details.
Loan Officer	Manages loan type records and borrower loan details.
Read	Provides read permission for all components.
Write	Provides write permission for all components.

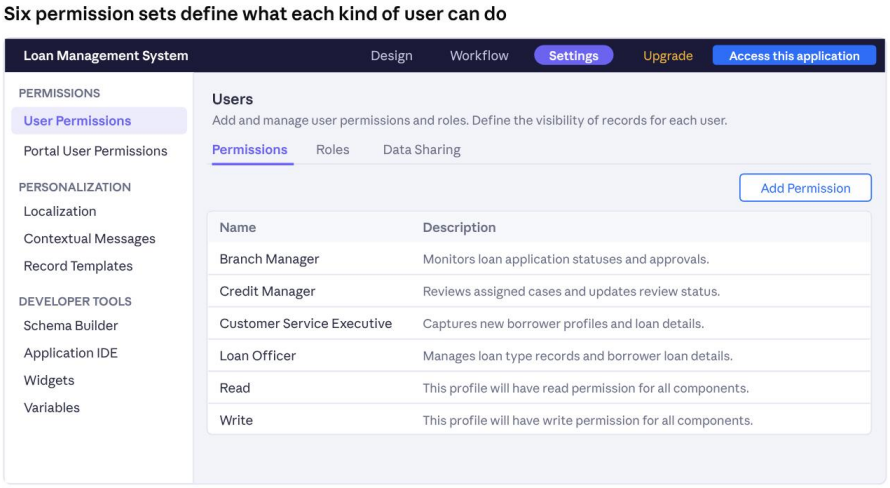


Figure 8.3: User Permissions - Permissions Section (Illustrative Representation, Not a Screenshot)

Step 5: Build the Role Hierarchy

On the Roles tab, the role tree is arranged with the CEO at the top. Branch Manager reports to the CEO, and Credit Manager, Loan Officer and Customer Service Executive all report to Branch

Manager. The Expand All, Collapse All, Manage Role Hierarchy and Add Role options are used as needed while building the tree.

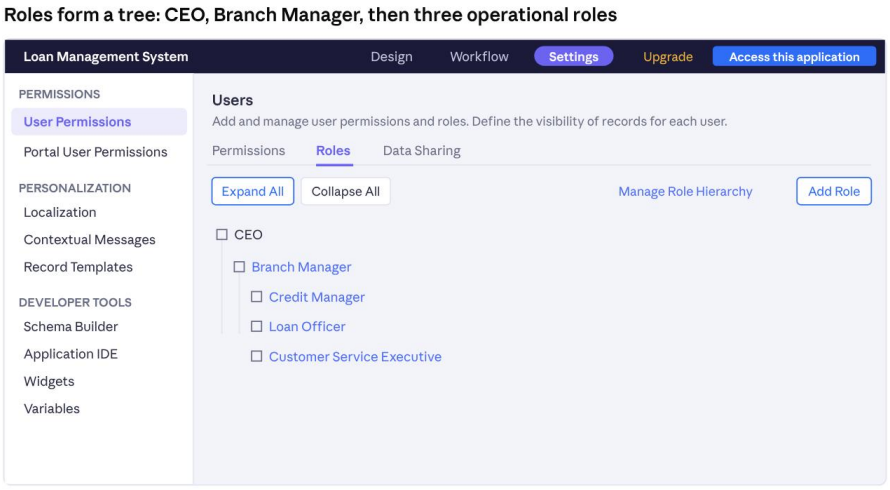


Figure 8.4: Roles and Role Hierarchy (Illustrative Representation, Not a Screenshot)

Figure 8.4 shows the completed hierarchy, in which the three operational roles sit side by side beneath Branch Manager as sibling roles.

Step 6: Create a New Share Rule

On the Data Sharing tab, New Share Rule is selected. The rule is named Customer_Service_to_Branch_Manager and the Form is set to Loan Application. Records shared from is set to Roles → Customer Service Executive, Records shared To is set to Roles → Branch Manager, and Level of access is set to Read/Write.

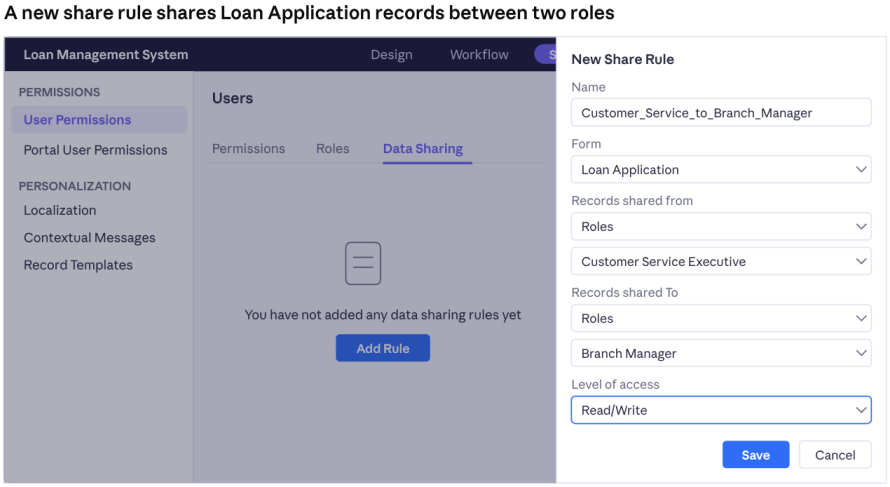


Figure 8.5: New Share Rule - Rule Configuration (Illustrative Representation, Not a Screenshot)

The New Share Rule panel in Figure 8.5 displays these values for the Loan Application form.

Step 7: Save and Verify the Rule

Selecting Save adds the rule to the Data Sharing list. The rule is then reopened with Edit Sharing Rule to confirm that every field was stored exactly as configured.

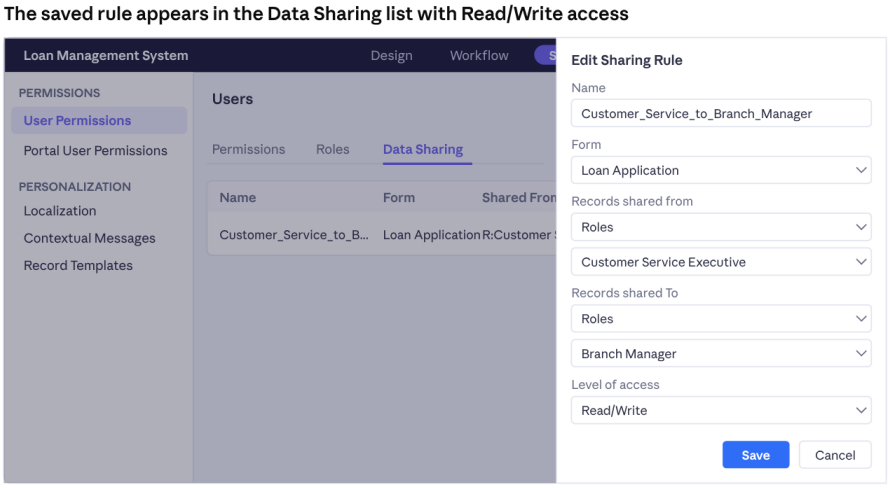


Figure 8.6: Configured Data Sharing Rule (Customer Service Executive → Branch Manager, Read/Write) (Illustrative Representation, Not a Screenshot)

The Data Sharing list now shows the saved rule with Form: Loan Application, Shared From: R:Customer Service Executive and Shared To: R:Branch Manager. The Edit Sharing Rule panel confirms the Read/Write access level.

8. Data Sharing and Configuration Details

8.1 Layered Access Model

The application follows a layered model: the forms (Borrower, Loan Application, Loan Type and Repayment) lead to the Dashboard, which is protected by a Security Layer made up of Permissions, Roles and Data Sharing. The role hierarchy runs from the CEO, who has overall oversight, down through Branch Manager to Credit Manager, Loan Officer and Customer Service Executive. As a result, each role sees the records of the roles beneath it.

8.2 The Configured Exception

The configured share rule adds one exception to this pattern. Loan applications captured by the Customer Service Executive role are shared with the Branch Manager role at Read/Write access, so that the branch can review and update them directly.

8.3 Rule Configuration Summary

Parameter	Detail
Rule Name	Customer_Service_to_Branch_Manager
Form	Loan Application
Records Shared From	Roles → Customer Service Executive
Records Shared To	Roles → Branch Manager
Level of Access	Read/Write

9. Result

- The Zoho Creator home page and the Loan Management System dashboard were reviewed as the entry points to the application.
- Six permission sets (Branch Manager, Credit Manager, Customer Service Executive, Loan Officer, Read and Write) were reviewed on the Permissions tab.
- The role hierarchy was arranged as CEO → Branch Manager → Credit Manager, Loan Officer and Customer Service Executive.
- A data sharing rule, Customer_Service_to_Branch_Manager, was created on the Loan Application form with Read/Write access, saved, and verified by reopening it.

10. Conclusion

This experiment showed how Zoho Creator controls access through three cooperating layers. Permission sets define what each kind of user may do, the role hierarchy decides whose records each role can see, and data sharing rules open selected records to other roles when the hierarchy alone is not sufficient. Configuring the rule through the form and verifying it by reopening it demonstrated that access can be set up without writing any code. This keeps loan data visible to the people who need it and hidden from everyone else.

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